

WHEELER AAF, HI, USA

WMO: 911700

Lat: 21.483N

Lon: 158.033W

Elev: 837

StdP: 14.26

Time Zone: -10.00 (W10)

Period: 06-19

WBAN: 22508

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	58.4	59.5	51.1	57.3	64.7	53.0	61.6	65.6	20.6	71.1	18.7	71.1	3.4	320	0.281

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.4	85.2	73.0	84.1	72.5	83.0	71.7	75.9	80.6	74.8	79.9	73.8	79.4	10.8	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.7	134.1	77.8	73.3	127.5	77.1	72.3	123.3	76.6	39.9	79.9	38.8	80.4	37.9	79.3	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	16.5	15.1	DB	54.5	87.0	1.0	1.7	53.7	88.3	53.1	89.3	52.5	90.3	51.8	91.6	(4)
(5)			WB	52.3	76.1	1.2	2.3	51.5	77.7	50.8	79.1	50.2	80.3	49.3	82.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.5	68.8	68.7	69.4	71.0	72.5	74.3	75.6	76.2	75.7	74.6	72.5	70.4	(6)	
	DBStd	3.32	2.11	2.38	2.33	2.25	1.74	1.59	1.91	1.84	1.79	1.70	2.08	2.00	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	4	1	2	1	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	8212	584	524	601	630	699	730	792	812	771	763	674	632	(10)	
	CDD65	2741	120	106	137	180	234	280	327	347	321	298	224	167	(11)	
	CDH74	12643	256	213	333	557	981	1438	1958	2232	2002	1525	804	344	(12)	
	CDH80	1555	1	1	5	13	53	150	313	418	363	191	42	5	(13)	
(14) Wind	WSAvg	6.2	5.6	6.1	6.6	6.5	6.2	6.7	6.7	6.6	6.0	5.7	5.7	5.6	(14)	
(15) Precipitation	PrecAvg	45.2	7.4	4.7	4.8	3.7	2.8	1.5	1.9	1.6	1.9	4.1	5.6	5.0	(15)	
	PrecMax	66.3	23.4	18.2	13.9	16.6	9.4	3.7	4.2	5.3	7.2	11.9	17.6	13.9	(16)	
	PrecMin	27.9	0.9	0.5	0.4	0.2	0.2	0.1	0.4	0.4	0.0	0.9	0.6	0.4	(17)	
	PrecStd	11.5	6.3	3.9	4.1	3.3	2.0	1.0	1.1	1.3	1.5	2.8	4.4	3.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.9	79.4	80.9	81.5	83.2	84.5	85.7	87.0	86.4	84.8	82.6	80.8	(19)	
		MCWB	68.2	68.7	69.2	68.9	71.0	71.7	73.1	73.9	73.5	73.6	72.3	69.6	(20)	
	2%	DB	78.1	77.9	79.0	80.2	81.4	82.8	84.2	85.2	84.8	83.4	81.3	78.9	(21)	
		MCWB	67.7	67.9	68.4	68.6	69.3	70.6	72.4	73.1	72.8	72.8	71.2	69.0	(22)	
	5%	DB	76.9	76.8	77.3	79.0	80.4	81.8	83.0	83.9	83.7	82.0	80.0	77.3	(23)	
		MCWB	67.3	67.4	67.3	68.2	69.0	69.6	71.6	72.4	72.3	71.6	70.5	68.4	(24)	
	10%	DB	75.2	75.1	75.6	77.4	79.1	80.8	81.9	82.3	82.1	81.1	78.7	76.0	(25)	
		MCWB	66.7	66.8	66.8	67.8	68.6	69.3	70.8	70.9	70.8	71.0	69.8	68.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.3	72.5	72.1	72.9	73.2	75.3	76.3	77.4	76.7	77.0	75.4	72.4	(27)	
		MCDB	75.8	75.1	77.2	76.2	79.6	78.3	80.9	82.7	81.7	80.7	79.1	76.1	(28)	
	2%	WB	70.1	70.5	70.4	71.1	71.8	73.0	74.4	75.5	75.4	75.1	73.5	71.4	(29)	
		MCDB	74.6	74.6	75.7	75.9	78.5	78.8	79.9	80.9	80.2	79.9	77.9	75.7	(30)	
	5%	WB	69.0	69.4	69.2	70.0	70.9	71.6	73.3	74.3	74.2	73.8	72.4	70.5	(31)	
		MCDB	73.8	73.8	74.4	75.3	77.5	78.9	79.6	80.2	79.9	79.2	77.2	74.6	(32)	
	10%	WB	68.1	68.3	68.0	69.1	70.0	70.8	72.3	73.1	73.1	72.9	71.3	69.6	(33)	
		MCDB	73.2	73.1	73.6	74.8	76.4	78.1	79.0	79.4	79.4	78.5	76.3	73.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.7	13.0	13.0	12.8	13.7	13.3	13.1	13.4	14.1	13.5	12.6	11.8	(35)	
		MCDBR	15.7	15.2	15.2	14.8	15.3	14.6	14.1	14.7	15.4	14.9	14.5	13.9	(36)	
	5% WB	MCWBR	7.9	7.7	7.1	6.6	6.6	5.6	5.3	5.6	6.2	6.5	6.8	6.8	(37)	
		MCWBR	13.8	12.9	14.0	12.9	14.0	13.6	12.7	13.4	12.8	13.6	13.1	12.0	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.344	0.350	0.357	0.374	0.368	0.357	0.356	0.355	0.348	0.358	0.356	0.340	(40)	
		taud	2.615	2.586	2.573	2.506	2.556	2.596	2.611	2.614	2.638	2.617	2.613	2.648	(41)	
	Ebn at Noon	Ebn at Noon	296	300	300	294	293	294	294	296	299	293	289	293	(42)	
		Edh at Noon	28	30	32	35	33	31	31	31	30	29	28	26	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1401	1578	1812	2045	2193	2294	2255	2164	2025	1701	1434	1295	(44)	
	RadStd	RadStd	60	113	192	115	125	63	46	86	61	55	38	92	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page